



FIRST SEMESTER 2026-27
COURSE HANDOUT Part II

Date: 02.08.2026

In addition to the part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : SNS F211
Course Title : Fundamentals of Nanoscience and Nanotechnology
Instructor-in-Charge : Prof. Sarbani Ghosh (sarbani.ghosh@pilani.bits-pilani.ac.in)

Class Timings : M W 5 Th 10 Hour

1. Course Description:

Nanotechnology is widely regarded as a transformative field with the potential to address critical challenges in energy, healthcare, electronics, and advanced manufacturing. This course is designed to introduce students to the fundamental concepts of nanoscience and nanotechnology, preparing them for further academic research or industrial careers in this rapidly evolving area. The course focuses on the fundamental science behind nanomaterials, their synthesis and properties, with emphasis on their size-dependent physical, chemical, electrical, optical, and thermal properties. Applications of nanomaterials in electronics, medicine, MEMS, photonics, catalysis, energy systems, and heat transfer will be discussed. The course also introduces essential quantum mechanical concepts relevant to nanoscale systems. In addition, emerging trends, large-scale manufacturing challenges, and future directions in nanoscience and nanotechnology are covered.

2. Scope and Objective of the Course:

The course is structured to provide a comprehensive and interdisciplinary overview of nanoscience and nanotechnology. Students will learn common methods for the synthesis of nanoparticles and nanostructures.

Special emphasis is placed on understanding the unique properties of nanomaterials and the scientific principles governing their behavior at the nanoscale. The course highlights real-world motivations and applications of nanomaterials across multiple technological domains.

This course is well suited for students seeking foundational knowledge and practical exposure to nanoscience and nanotechnology, and serves as an effective entry point into advanced study or research in the field.

3. Course Learning Outcomes:

After successful completion of this course, the student will be able to:



- Explain the fundamental principles of nanoscience, including size-dependent properties and quantum effects at the nanoscale.
- Describe and compare common synthesis techniques used for nanomaterials.
- Analyze the structure–property–application relationships of nanomaterials in areas such as electronics, energy, catalysis, and biomedical systems.

4. Text Books:

TB-1: William A. Goddard, III, Donald W. Brenner, Sergey E. Lyshevski and Gerald J. Iafrate, "Handbook of Nanoscience, Engineering and Technology", CRC Press, ed 3 (March 12, 2012) ISBN-13: 978-1- 4398-6016-8.

TB-2: Risal Singh and Shipra Mital Gupta, "Introduction to Nanotechnology", Oxford University Press, ed 1 (2016). ISBN-13:978-0-19-945678-9

TB-3: James Murday, B S Murty, P Shankar, Baldev Raj, and BB Rath, "Textbook of Nanoscience and Nanotechnology" University Press – IIM; ed 1 (2012) ISBN: 978-81-7371-738-3.

5. Reference Books:

1. Fundamentals of Nanoscience, S L Kakani & Shubhra Kakani, New Age International Publishers
2. Bharat Bhushan, "Handbook of Nanotechnology", Springer, Germany, 2nd ed., 2006, ISBN 13: 9783540298557
3. Hari Singh Nalwa, "Handbook of Nanostructured Materials and Nanotechnology", Academic Press, 2001, ISBN 13: 9780080533643.
4. Paul Harrison, "Quantum Wells, Wires & Dots, Theoretical & Computational Physics of Semiconductors Nanostructures", John Wiley & Sons, Ltd, 2nd ed., 2006, ISBN 13: 9780470010792.
5. Charles P Poole & Frank J. Ownes, "Introduction to Nanotechnology", Wiley-Interscience; 1st ed., ISBN-13: 978-0471079354
6. T. Pradeep, "A textbook of Nanoscience & Nanotechnology," McGraw Hill India, 1st ed., 2015, ISBN 13: 9781259007323.

6. Course Plan:

Module No/ Module Name	Topics to be covered	Learning Outcomes	Chapter in book
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1 Introduction	L1-L10: Course introduction, what is nanoscience and nanotechnology? Difference between micro and nano, classification of nanostructured materials, unique properties of nanomaterials: structural, electronic, magnetic, optical, mechanical.	- Elaborate the difference between nano and microscale - List out the key properties of nanomaterials.	TB-1+TB-2 + TB-3 + Notes in class
2 Nanomaterials Synthesis Techniques: Bottom-up	L11-L14: Bottom-up: physical vapor deposition (PVD), chemical vapor deposition (CVD), sol- gel, spray conversion, wet chemical synthesis, self-assembly	Discuss various bottom- up techniques for nanomaterials synthesis	TB-1+TB-2 + TB-3 + Notes in class
3 Nanomaterials Synthesis Techniques: Top-down	L15-L17: Top-Down: Mechanical alloying, equal channel angular pressing, high-pressure torsion (HPT), accumulative roll bonding (ARB), Nanolithography	Discuss various top-down approaches for nanomaterials synthesis	TB-1+TB-2 + TB-3 + Notes in class
4 Quantum Mechanical Aspects of Nanomaterials	L18-L20: Basics of quantum physics, wave mechanics, Schrödinger wave equation, concepts of quantum well, wire and dots, tunneling, applications of quantum mechanics in nanotechnology.	Elaborate the basic physics behind quantum effects exhibited by nanomaterials.	TB-2+ Notes in class
5 Carbon-based Nanostructures	L21-L23: Synthesis, properties, and applications of carbon nanostructures: fullerenes, carbon nanotubes (CNTs), graphene, nanodiamonds, and diamond-like carbon (DLC).	Identify various nano- structured forms of carbon and their synthesis and unique properties.	TB-1+TB-2 + TB-3 + Notes in class



6 Nanotechnology Applications: Electronics	L24-L25: Applications of nanotechnology in electronics: Basics of semiconductor devices, MOSFETs, FinFETs	Review and suggest nanotechnology applications in electronic devices	Discuss in class from research papers
7 Nanotechnology Applications: Medical	L26-L28: Application of nanotechnology in medical devices: Basics of bio-electronic devices, nanosensors, molecular imaging, targeted drug delivery applications.	Review and suggest nanotechnology applications in medical devices	Discuss in class from research papers
8 Nanotechnology Applications: MEMS/NEMS	L29-L31: Application of nanotechnology in MEMS: Basics of MEMS/NEMS, fabrication techniques, working principles of accelerometers, and actuators.	Review and suggest nanotechnology applications in MEMS devices	Discuss in class from research papers
9 Nanotechnology Applications: Energy	L32-L34: Basics of fuel cells and batteries, nanotechnology applications in fuel cells, energy storage devices.	Review and suggest nanotechnology applications in energy storage.	Discuss in class from research papers
10 Nanotechnology Applications: Catalysis	L35-L37: Basics of catalysis, synthesis of metal and oxide active phases, nanoreactors.	Review and suggest nanotechnology applications in catalysis	Discuss in class from research papers
11 Nanotechnology Applications: Heat Transfer	L38-L39: Nanofluids Synthesis, characterization and applications relevant to heat transfer. State of the art and future research directions.	Review and suggest nanotechnology applications in heat transfer	Discuss in class from research papers
12		Explain the hurdles	Discuss in



Nanotech hurdles and challenges	L40: Large scale manufacture and technological issues	and avenues for further research in the field of nanotechnology.	class from research papers
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7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Surprised Quiz (best ~3 out of 4)	20 min	10	During Class Hours	CB
Tutorial Tests (5 out of 6) Announced	20 min	25	During Tutorial Class Hours (TBA)	CB
Midsemester	90 min	25	07/10/2026	CB and/or OB
Comprehensive	180 min	40	08/12/2026	CB and/or OB

8. Online/Chamber Consultations: TBA.

9. Notices: All Notices would be put up on Google Classroom/Nalanda and/or be announced during the class. All students attending this course MUST join google classroom with the class ID **TBA**. All course related notices and announcements will be posted in the classroom. Assignments may also be posted in classroom.

10. Make-up Policy: Make-up will be granted for **genuine cases only** where hospitalization is prescribed. Certificate from an CMO must accompany the make-up application. Except for an emergency, for all other cases, **prior permission of IC is compulsory**.

No make-up for class/tutorial tests and assignments.

11. Note (if any): *The lecture notes (if any) provided to the students registered in this course is for academic purpose only and is considered the intellectual property of BITS Pilani. Such materials should not be uploaded to any external websites/forums without the permission of the Instructor.*

Sarbani Ghosh
Instructor-in-charge
Course No. BITS F416